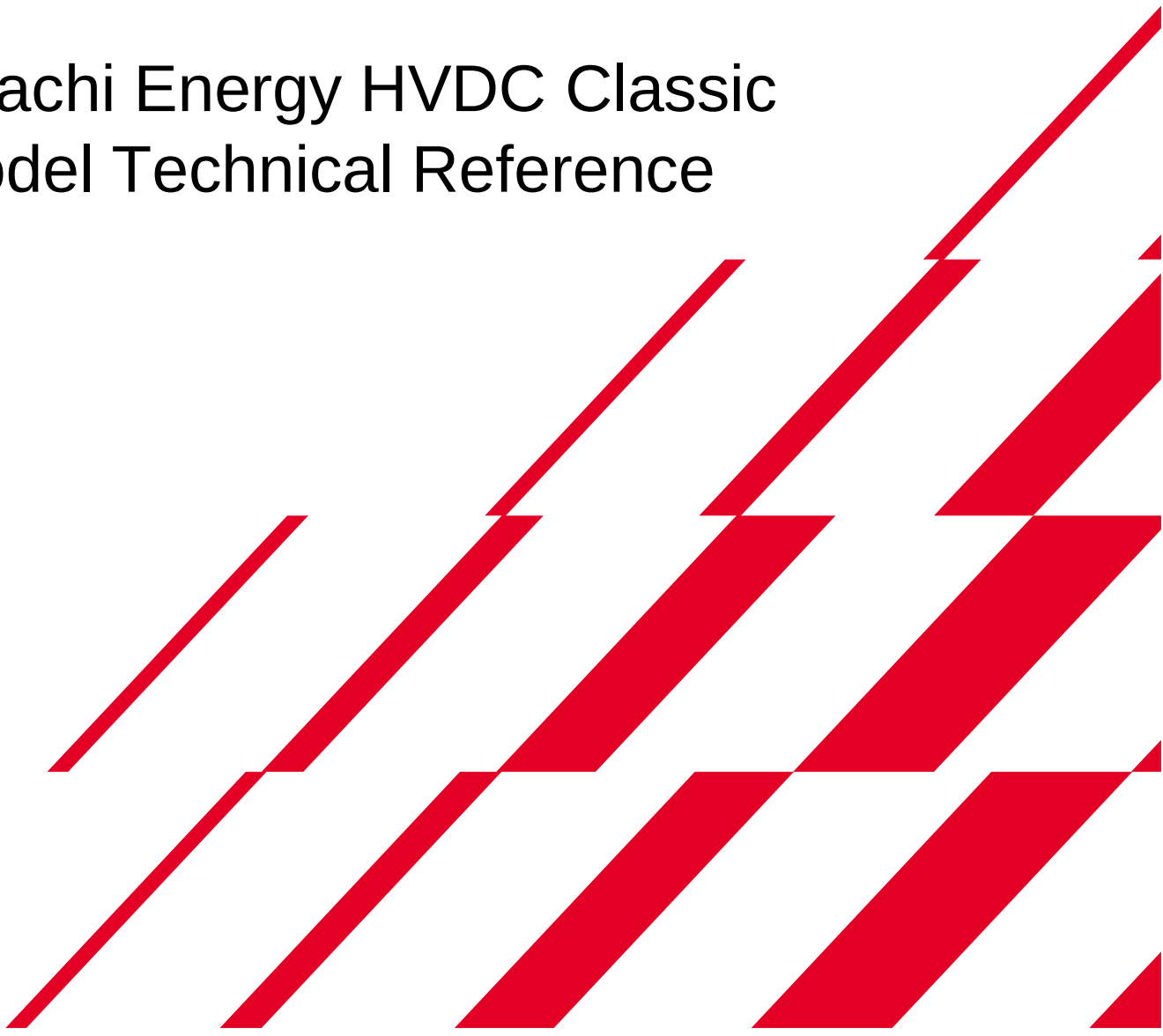
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Hitachi Energy HVDC Classic Model Technical Reference



Hitachi Energy

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1. Overview

This document gives instructions on how to set up the load flow, solve the load flow, run the dynamic simulation, and operate the Hitachi Energy HVDC Classic library model for PowerFactory. The model is limited to fundamental frequency AC quantities, described by phasors, and mean values of DC quantities.

The model consists of an end-to-end representation of the HVDC system, including converter stations, DC system, and AC network equivalents. The model can be used for load flow calculations and dynamic simulation.

The electrical circuit of the HVDC converters is built up from the standard elements available in the PowerFactory program. The dynamic HVDC Classic model representing the dynamic performance of the HVDC Classic transmission is a Dynamic Linked Library, DLL, and linked into PowerFactory.

The DC side electrical representation is defined within PowerFactory itself by its standard components.

2. Load flow representation

This section provides a detailed description of the load flow model including instructions on how to use it. Representation in the form described below is required for successful initialization of the dynamic model and the given rules must be followed to achieve proper function of the control system.

2.1 Converter representation

The converter transformer and bridge are represented with the standard PowerFactory Rectifier/Inverter 2-DC connection (ElmRec) component in the electrical circuit.

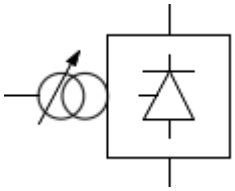


Figure 1: ElmRec component in PowerFactory.

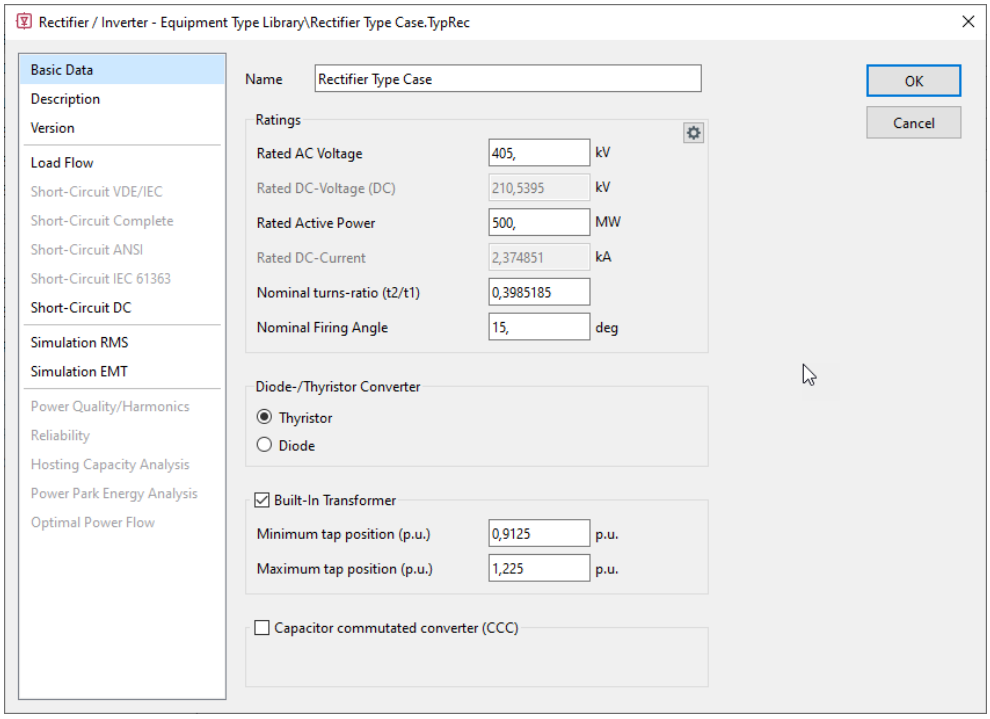


Figure 2: HVDC converter type.

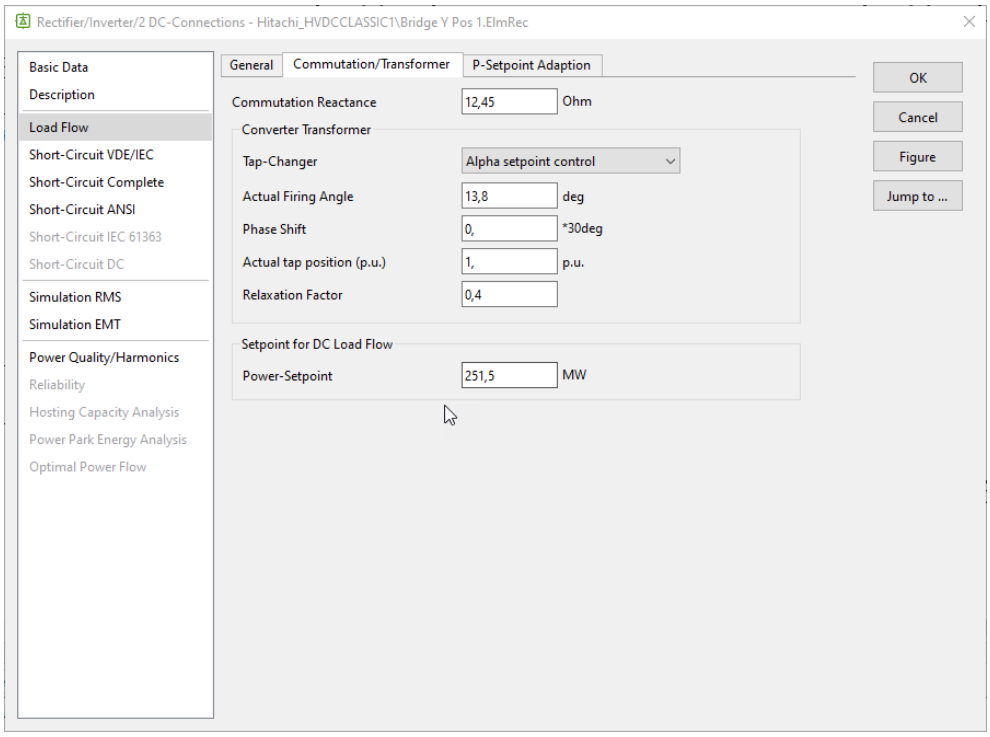


Figure 3: HVDC converter element.

The following parameters are valid for the HVDC Classic converter, PowerFactory ElmRec component.

Table 1: Parameters of the HVDC converter type.

Parameter	Description	Unit	Used in dynamic model
Unom	Rated AC Voltage	kV	Not used in dynamic model
Unomdc	Rated DC Voltage	kV	Not used in dynamic model
Pnom	Rated Active Power	MW	Not used in dynamic model
Imax	Rated DC Current	kA	Not used in dynamic model
tapnom	Nominal Turns-Ratio (t2/t1)	-	Yes, it needs to be the same as the ‘trr’ value used in the dynamic model parameters
alphanom	Nominal Firing Angle	deg	Not used in dynamic model
tapmin	Converter Transformer: Minimum Turns-Ratio	pu	Not used in dynamic model
tapmax	Converter Transformer: Maximum Turns-Ratio	pu	Not used in dynamic model
Pnold	No Load losses	kW	Not used in dynamic model
swtLossFactor	Switching loss factor	kW/A	Not used in dynamic model
resLossFactor	Resistive loss factor	Ohm	Not used in dynamic model

Table 2: Parameters of the HVDC converter element.

Parameter	Description	Unit	Used in dynamic model
bstp	Firing Angle (alpha) Control: Control Characteristic	-	Yes, it needs to be followed by the definitions in Table 3 for rectifier or inverter
alphamin	Firing Angle (alpha) Control: Minimum Firing Angle	deg	Not used in dynamic model
alphamax	Firing Angle (alpha) Control: Maximum Firing Angle	deg	Not used in dynamic model
gammamin	Firing Angle (alpha) Control: Minimum Extinction Angle	deg	Not used in dynamic model
Xd	Converter Transformer: Commutation Reactance	Ohm	Not used in dynamic model
nt2ag	Converter Transformer: Phase Shift	*30deg	Not used in dynamic model
alpha set	Firing Angle (alpha) Control: Actual Firing-Angle	deg	Not used in dynamic model
nntap	Converter Transformer: Actual Winding Ratio	pu	Not used in dynamic model
Pset	Firing Angle (alpha) Control: Power-Setpoint	MW	Yes, it needs to be the same as the ‘Order’ value used in the dynamic model parameters from HLC inputs Note: Order = Bridge1 + Bridge2
uset	Firing Angle (alpha) Control: Voltage Setpoint	pu	Yes, it needs to be the same as the ‘UdRefkV’ value used in the dynamic model parameters from HLC inputs

Since the 12-pulse bridge converter is a default module to get a convergence load flow, the 6-pulse bridge converter groups in each station must be properly coordinated.

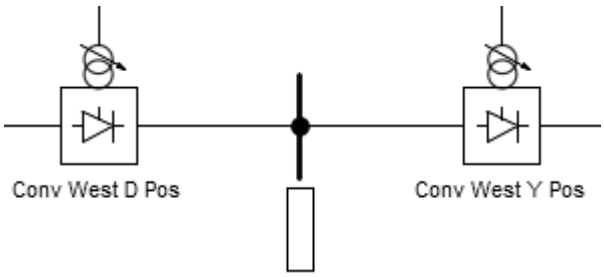


Figure 4: 12 pulses bridge in PowerFactory electrical circuit.

To solve the load flow more efficiently, the following setup for ElmRec is recommended:

Table 3: Control/ tap changer characteristics.

Name	Control-Characteristic	Tap-Changer
Rectifier, Y Group	P(Power control)	Alpha-control
Rectifier, D Group	P(Power control)	Alpha-control
Inverter, Y Group	Vdc(Voltage control)	Gamma-control
Inverter, D Group	EXT(External control, assign the controller to the nearest Y group, i.e. the same firing alpha is used within this 12 pulses bridge group)	Gamma-control

2.2 DC system

The DC system, which includes the smoothing reactor, DC filter, and cable/line is presented in an electrical circuit by individual predefined components.

2.3 Configurations

The model is a typical monopolar configuration with a cable transmission line with one pole and ground return.

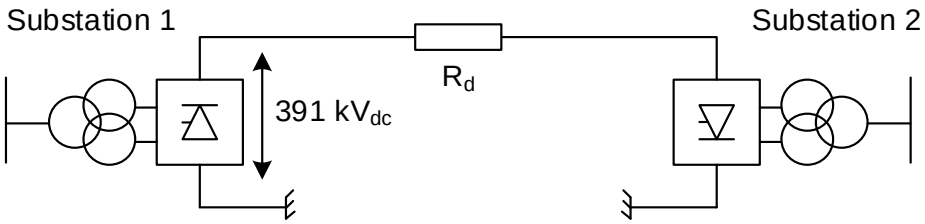


Figure 5: Monopolar configuration.

3. Dynamic Model

This section summarizes how the dynamic data of the model is defined and how the model is used. The input signals are measured from various points in the grid and passed on to the control system. The control system processes the input signals and passes on the output signals, which is the firing angle alpha, to the block named 6-Pulse Bridge Rectifier/Inverter, which is then linked to the Bridge (ElmRec) used in the grid.

The model is suitable for the time integration step in the range of 0.5ms to 10ms. The control system runs internally with an internal time step of 0.5ms independent of the time step used in PowerFactory. For studies using weak AC networks, a time step in the range of 0.5ms to 1ms is recommended.

Time step adaptation is also possible, the recommendation is to use the electromechanical transients in 0.5ms and the maximum time step size equal to 10ms.

3.1 Interactions between code and electrical circuit

The dynamic behavior of the converter control is modeled by the external model interacting with the electrical circuit representation in PowerFactory as shown in Figure 6.

During the dynamic simulation, measurement signals and control parameter inputs are sent from the PowerFactory to the external converter control model, and the firing orders for each converter bridge are calculated externally and sent back to the converter bridge components on the PowerFactory circuit diagram. By using the firing alpha order from external control, PowerFactory builds up the DC terminal voltage internally, and together with the DC side electrical representation (Line/Cable), the dynamic behavior of the DC transmission is modeled.

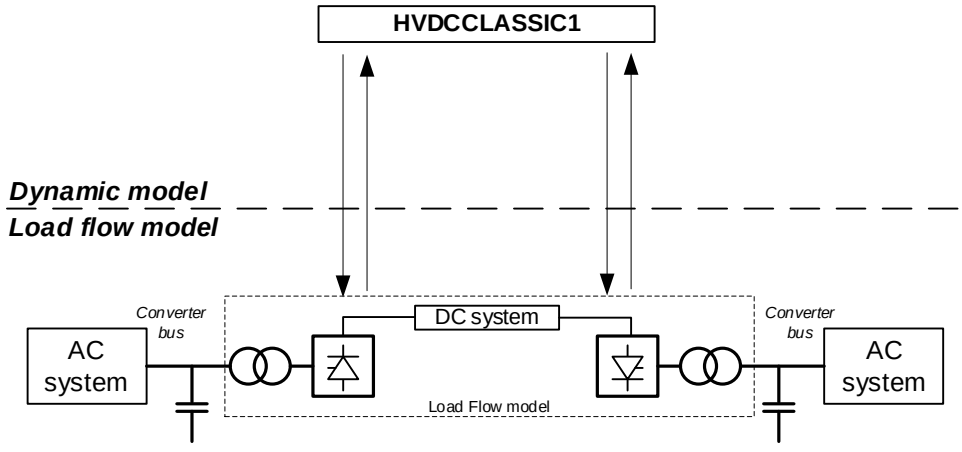


Figure 6: Overview of the interaction of the dynamic model.

3.2 Converter station frame

The slots in the converter station frame are defined in Table 4.

Table 4: Slots in the converter station.

Slot name	Type of element	Description
Ctrl_refs_conv1	ElmDsl	Provides external step in the reference values to control 1
Ctrl_modes_conv1	ElmDsl	Provides external parameters to control 1
Time	ElmDsl	Provides the current time to both controls
Uac_meas_Conv1	StaVmea	Measures the AC bus voltage from station 1 side
Udc_meas_Conv1	StaVmea	Measures the DC voltage from station 1 side
Idc_meas_Conv1	StaImea	Measures the DC current from the station 1 side
Tap Changer Controller1	ElmDsl	Reads the transformer tap position from the station 1 side
Telecom_1	ElmDsl	Telecommunication between the converters
HLC_inputs	ElmDsl	Provides reference values to both controls and optional modulation signal for the Active power controller
Init_Warn1	ElmDsl	Detects and displays the initialization warnings
Conv1 Y Bridge	ElmRec	Measurements and link the control frame outputs to the converter bridge Y Pos 1 (ElmRec) used in the grid
Conv1 D Bridge	ElmRec	Measurements and link the control frame outputs to the converter bridge D Pos 1 (ElmRec) used in the grid
HVDC Control Conv1	ElmMdl	Calls the external model library file (DLL) for control 1
Ctrl_refs_conv2	ElmDsl	Provides external step in the reference values to control 2
Ctrl_modes_conv2	ElmDsl	Provides external parameters to control 2
Uac_meas_Conv2	StaVmea	Measures the AC bus voltage from station 2 side
Udc_meas_Conv2	StaVmea	Measures the DC voltage from station 2 side
Idc_meas_Conv2	StaImea	Measures the DC current from the station 2 side
Tap Changer Controller2	ElmDsl	Reads the transformer tap position from station 2 side
Telecom_2	ElmDsl	Telecommunication between the converters
Init_Warn2	ElmDsl	Detects and displays the initialization warnings
Conv2 Y Bridge	ElmRec	Measurements and link the control frame outputs to the converter bridge Y Pos 2 (ElmRec) used in the grid
Conv2 D Bridge	ElmRec	Measurements and link the control frame outputs to the converter bridge D Pos 2 (ElmRec) used in the grid
HVDC Control Conv2	ElmMdl	Calls the external model library file (DLL) for control 2

3.3 Control input values

Input values are the inputs to the control block. They are either measurement values or control signals that can be altered through user input (event change).

Table 5: Input values.

Signal name	Unit	Description
IoStep	A	Current order step
PoStep	MW	Power order step
UdRefStep	kV	DC voltage reference step
GammaRefStep	deg	Gamma angle reference step
Lead	1/0	1= Lead Station, 0= Trail
Rectifier	1/0	Operation mode for converter, 1=Rectifier, 0=Inverter

CurrDir	1/-1	Current direction for converter which will influence DC voltage polarity. Rectifier(1) CurrDir(1) 1 1 = Pos DC voltage 1 -1 = Neg DC voltage 0 1 = Neg DC voltage 0 -1 = Pos DC voltage
ExternTrigComFail	1/0	External trig of commutation failure, 0=Normal value, 1=Initiate
ManualCtrlAct	0-255	Manual Control (or Auxiliary) Action 0= No action 1= To be used as DC line ground fault detection, retard by fast recovery when released 2= To be used at auto reclose sequence, retard followed recovery when released 4= To be used before a block action sequence, retard, a fast recovery 8= Temporary (short time) block combined with retard o 16= A temporary (short time) block can be applied on one converter only. 32= To be used at ordered alpha90 operation, for instanc consume reactive power only or during inverter stop 64= To be used when an error is detected at DC voltage measurement, for instance, due to measuring-equipr damage 128= To be used to short time increase gamma order, to s overvoltage at switching of the filter bank.
Inspect	1/0	Log file activated, 0=No log-print, 1= log-print active
Rcol_Enable	1/0	Enable/Disable
Gamma0	1/0	Enable/Disable
Raml	1/0	Enable/Disable
IncreaseGamma	1/0	Enable/Disable
CfPrevInhibt	1/0	Enable/Disable
Internal 1 - 2	-	Internal use
Time	sec	Current Time
Uac_r	kV	AC bus voltage real part
Uac_i	kV	AC bus voltage imaginary part
Udc	kV	DC voltage
Idc	A	DC current
tapp	pu	Tap position
Unom	kV	Nominal voltage AC
frnom	Hz	Nominal frequency
Xd	Ohm	Commutation reactance
alpha_deg	deg	Firing angle alpha
gamma	rad	Firing angle gamma
Order_to_Pole	MW or A	Power order
CtrlMode_to_Pole	0-2	Control Mode, 0=Blocked. 1=Power Control, 2= Current Control
UdRefkV_to_Pole	kV	Scheduled DC voltage
DeltaP	MW	Optional power modulation input
Io70FoSta	A	Current order from other station

3.4 Control parameter values

Parameter values are the parameters of the control model. These values can be changed in the PowerFactory ElmMdl parameter dialog.

Table 6: Parameter values.

Parameter	Default	Min	Max	Unit	Description
ConvNr	1	0	99	1-99	Unique converter number
Rdc	8.9401	0	999	Ohm	DC resistance
Trr	0.398519	0	1	pu	Transformer Ratio
UdNom	391	0	3000	kV	Norm DC voltage, 12-pulse value
IdNom	1279	0	10000	A	Norm DC current
Udi0Nom	0	0	3000	kV	Norm Udi0 voltage
Gamma0Set	0.6	0	999	pu	Gamma0 control parameter
Gamma0Reset	0.75	0	999	pu	Gamma0 control parameter
UdTcUpRe	0.04	0	999	pu	VdCol control parameter
UdTcDnRe	0.015	0	999	pu	VdCol control parameter
UdTcUpInv	0.05	0	999	pu	VdCol control parameter
UdTcDnInv	0.015	0	999	pu	VdCol control parameter
IoAbsMax	2	0	999	pu	VdCol control parameter
IoAbsMin	0.1	0	999	pu	VdCol control parameter
IoLim	0.345	0	999	pu	VdCol control parameter
UdLowParamRe	0.25	0	999	pu	VdCol control parameter
UdLowParamInv	0.25	0	999	pu	VdCol control parameter
UdHigh	0.7	0	999	pu	VdCol control parameter
UdApcTfilt	1	0	999	sec	Ud calculation
MinUdRef	0.8	0	999	pu	System parameter
GammakickContr	5	0	999	deg	System parameter
GammaMin	12	0	999	deg	System parameter
AmaxGain	0.15	0	999	pu	AMAX control parameter
RamlRef	0.75	0	999	pu	RAML control parameter
CRamlRef	0.9	0	999	pu	RAML control parameter
DlTimeout	0.6	0	999	sec	RAML control parameter
DlLevel	30	0	999	deg	RAML control parameter
CDlTimeout	0.6	0	999	sec	RAML control parameter
CDlLevel	30	0	999	deg	RAML control parameter
RamlDecr	0.5	0	999	deg/ms	RAML control parameter
VcaGainP	100	0	9999	pu	Vca control parameter
VcaGainI	0.0006	0	9999	sec	Vca control parameter
CcaKp	1.8	0	9999	pu	Cca control parameter
CcaGain	40	0	9999	pu	Cca control parameter
CcaTi	0.01	0	9999	pu	Cca control parameter
Internal1 to 18	0	-	-	-	Internal use
IoStep	0	0	9999	A	Current order step

PoStep	0	-9999	9999	MW	Power order step
UdRefStep	0	0	9999	kV	DC voltage reference step
GammaRefStep	0	0	180	Deg	Gamma angle reference step

3.5 Control output values

Output values are used for traditional output, i.e., to be used to control 6-pulses bridges in the electrical system (converter representation) or for plotting purposes.

Table 7: Output values.

Parameter	Unit	Description
IoStepOut	A	Current order step, used for plotting
PoStepOut	MW	Power order step, used for plotting
UdRefStepOut	kV	DC voltage reference step, used for plotting
GammaRefStepOut	deg	Gamma angle reference step, used for plotting
InitWarn	-	Activate initialization warning
AlphaExecY	rad	Angle alpha executed to the bridge Y
AlphaExecD	rad	Angle alpha executed to the bridge D
Delta_F	Hz	Delta frequency
IordLim	pu	Current order limited
AlphaOrder	deg	Angle alpha order
Alpha	deg	Angle alpha
Gamma	deg	Angle gamma
Ud_Internal	kV	DC voltage
io_tosta	A	Current order to the other station

3.6 High-level control (HLC)

Parameter values from HLC are the inputs to the control block. They are reference signals that need to be synchronized with the electrical circuit load flow calculation and can be altered through user input (event change).

Table 8: HLC control parameters.

Parameter	Unit	Description
CtrlMode	0/1/2	0=Blocked. 1=Power Control, 2= Current Control
Order	A or MW	CtrlMode=1--power order CtrlMode=2--current order The total power order for the HVDC link must be synchronized to the values typed in the electrical circuit
UdRefkV	kV	Scheduled DC voltage must be synchronized to the values typed in the electrical circuit
Order_Factor	pu	Order distribute power between poles

3.7 Telecommunication

The parameter value from the Telecom block is used to define the telecommunication delay between stations.

Table 9: Telecom parameter.

Parameter	Unit	Description
TeleDelay	sec	Telecom communication delay between the two stations

3.8 Error message

The control system prints error messages to a PowerFactory output window. It is of great importance to check this when starting a simulation, to make sure that the model is set up correctly.

Error messages are also printed to a file on the system path for temporary files. The system path is defined by the system variable “temp” which can be shown by the Windows command:
echo %temp%

This command can be executed in a Windows command window, started by typing cmd in the Windows search window. Typically, this path is:

- C:\Users\AABBCC\AppData\Local\Temp, where AABBCC is the user's name.
- C:\Temp
- If not, %temp% is defined on the PowerFactory installation path.

The file printed is:

- PowerFactory_HVDC_HVDCCLASSIC1_1_ConvNo_1_Error.txt

It is important that the user has written permission to the used system path for temporary files, and that the path exists.

3.9 HVDC Classic model interface

The interface in PowerFactory to the external DLL is described in this section. The relationship between different elements in the HVDC Classic model is shown in Figure 7. There are different measuring objects and objects with inputs that are connected to the control block. The control block calls the DLL which contains the HVDC control system. The output signals from the control system, which are the alpha angle for both 6-pulse bridges, are then passed on to the block named Rectifier/Inverter, which is linked to the 6-pulse Bridge Converter used in the grid.

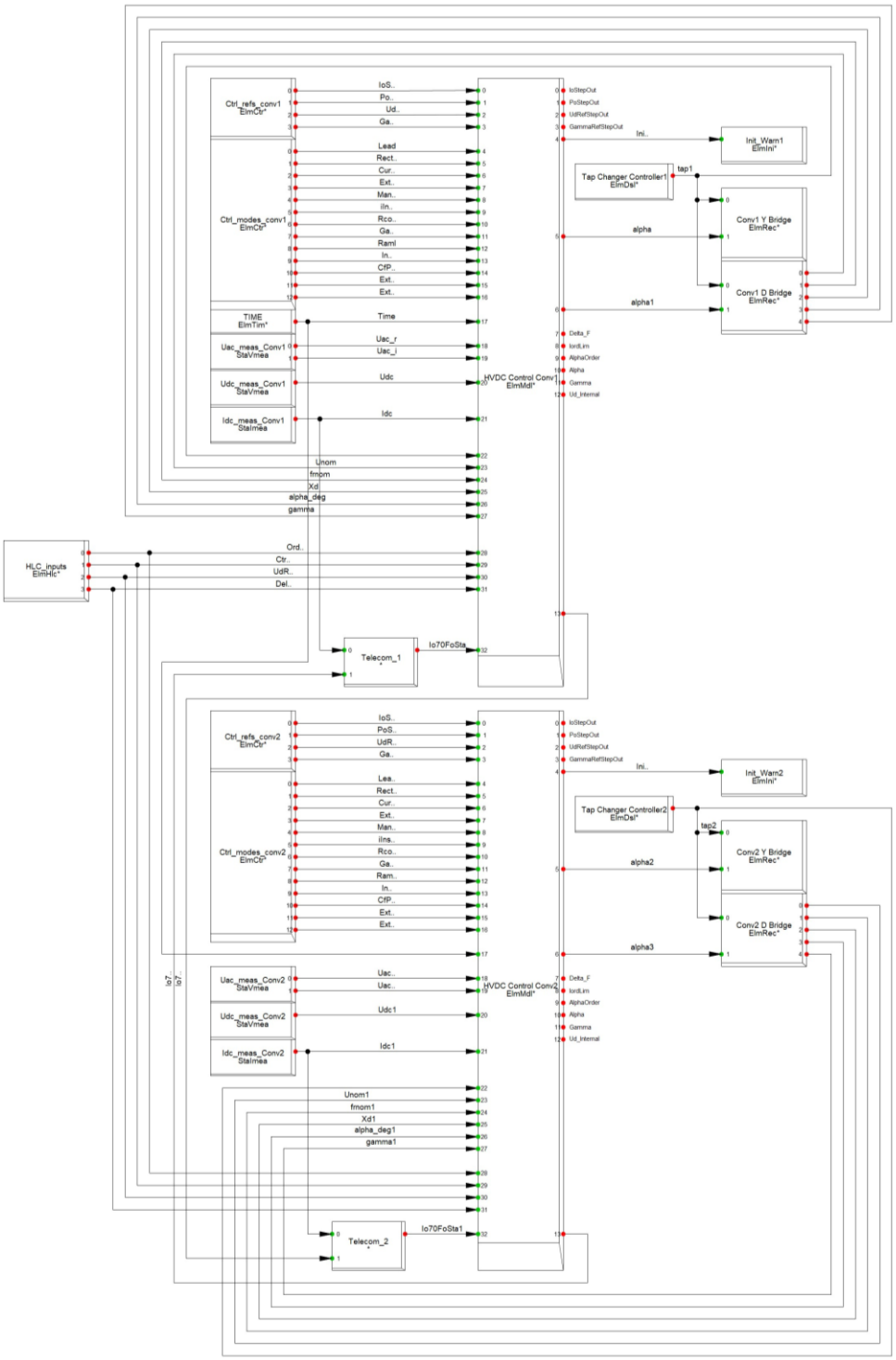


Figure 7: Overview of the control within PowerFactory.

Measurements, Control, and Converter slots in Figure 7 are declared and referenced to the specific object in the Composite model shown in Figure 8.

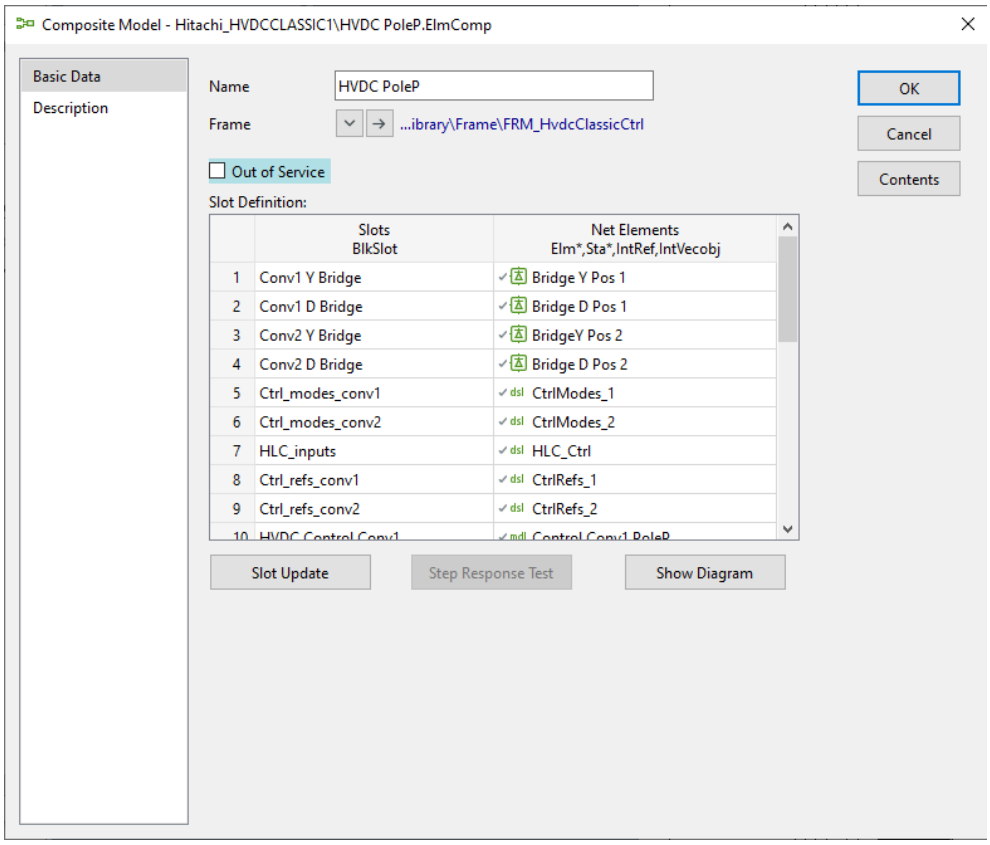


Figure 8: Control input interface.

Figure 9 shows the input parameters for the HVDC control system.

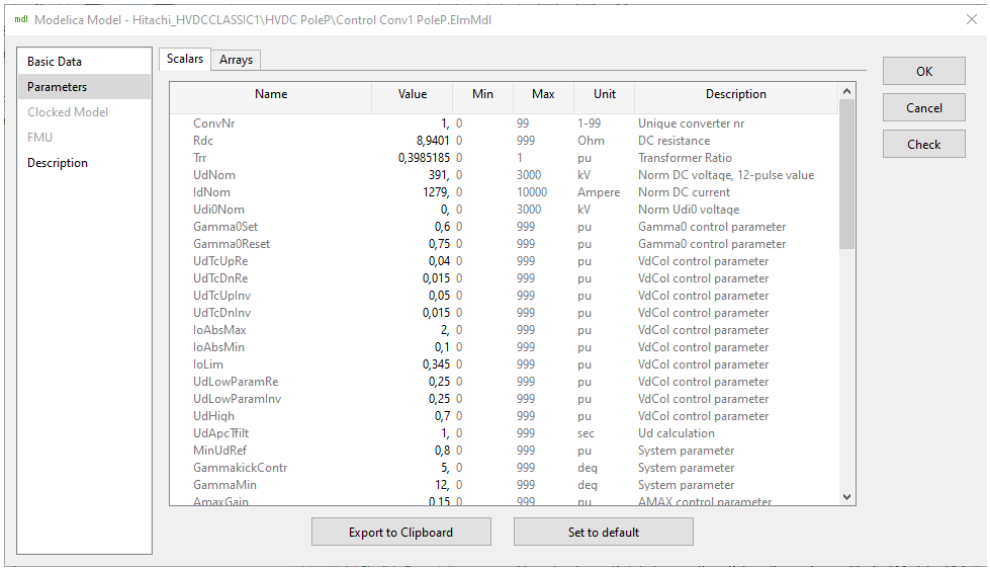


Figure 9: Parameter Interface of the control system.

The call to the external DLL is created in the Block Definition of the Common Model shown in Figure 10.

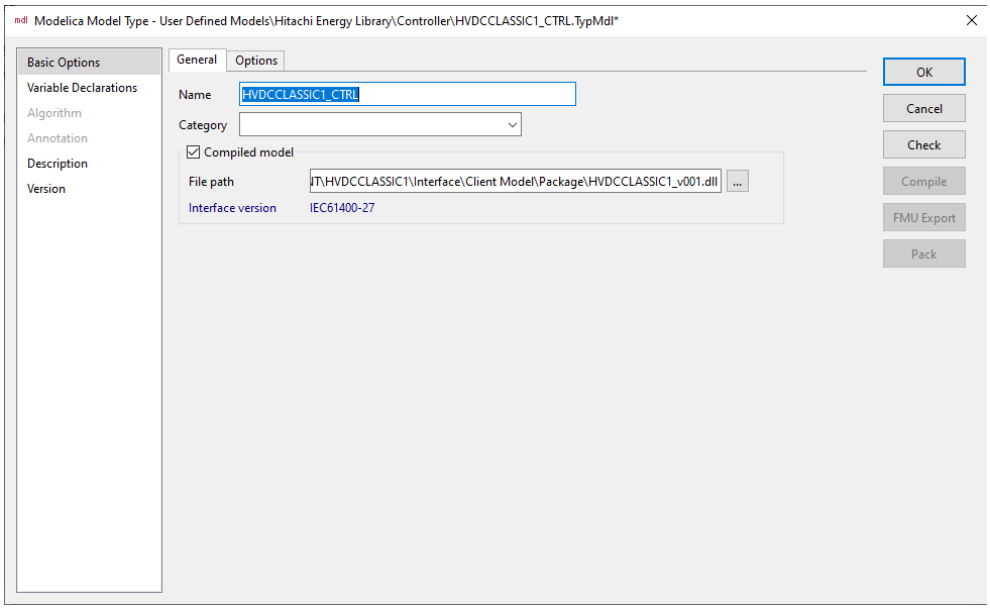


Figure 10: Block definition of the Common Model.

In this Block Definition, input signal and output signals are transferred between PowerFactory and the DLL.

3.10 Converter control

The principles of the converter control represented in the model are given in Figure 11.

The model includes the following control representation:

- APC, Active Power Control, calculates the current order in Power control, including current order synchronization between the two stations, and the telecom delay.
- CFC, Converter Firing Control.

  CFC includes the fundamental controls, and complementary transient controls, such as (but not limited to):

  - VDCOL, Voltage-Dependent Current Order.
  - CCA, Current Control Amplifier.
  - FirLim, Firing angle (Delta Alpha) limiter.
  - NewFir, Examine firing conditions.
  - AMAX, Alpha Maximum limitations for rectifier and inverter control.
  - Gamma0, inverter-specific control for faster recovery after a fault.
  - VcaReg, DC-voltage controller.
  - RAML, Rectifier Alpha Minimum Limiter, for smoother recovery after rectifier AC fault.
  - EMG, Emergency firing control.
  - UMIN, Minimum AC voltage firing control.
  - AMIN, Area min control, for inverter commutation margin control.
  - FMD, Firing Mode Decoder, defines symmetric firing conditions.
  - PLL, Phase Locked Loop (dynamic behavior).

- PCO, Phase Controlled Oscillator, internal definition of actual firing angle.

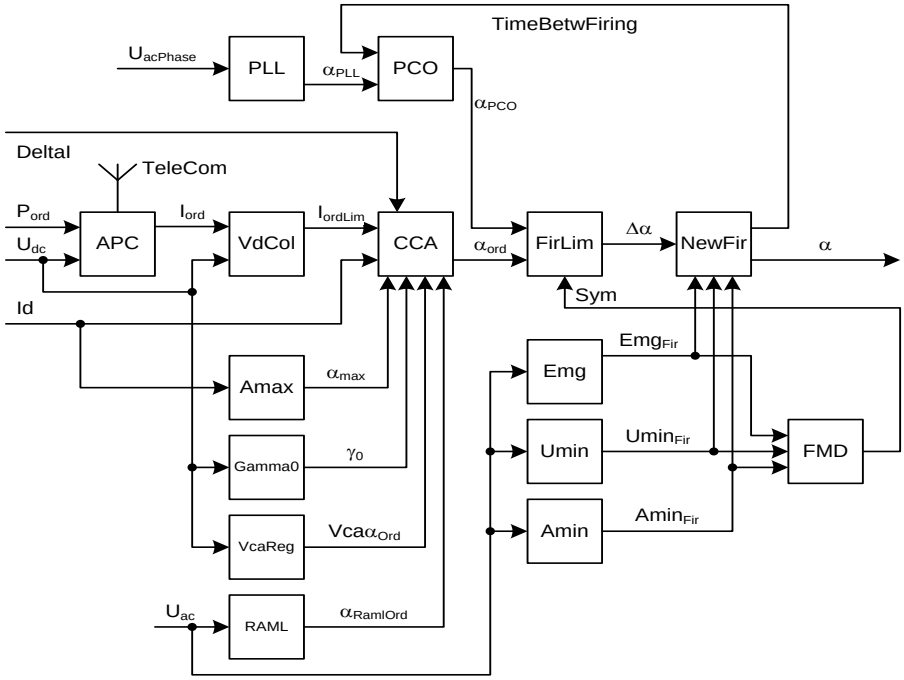


Figure 11: Principles of control system.

4. References

[1] DIgSILENT Technical Documentation, 6-Pulse Bridge

[2] https://www.hitachienergy.com/hvdc

Hitachi Energy
Uno Lamm HVDC Center SE-771 80
Ludvika Sweden
Tel: +46 21 32 50 00

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